	Recommended process*:				Desired functionality:					Typical product properties:				
o=standard ●=good ●●=very good ●●●=superior	Wet granulation	Dry granulation	Direct Compression	Capsule filling	Flow	Compaction	Content uniformity	Disintegration	Low moisture	Bulk density (g/mL)	Tapped density (g/mL)	x10 (µm)	x50 (µm)	x90 (µm)
Product														
Micronized lactose														
Lactochem® microfine	●●	o	o	o	o	●●●	●●●	●	●	0.28	0.43	0	3	7
Milled														
Pharmatose® 450M	●●●	●	o	o	o	●●●	●●●	●	●	0.50	0.74	3	20	50
Pharmatose® 350M	●●●	●	o	o	o	●●●	●●	●	●	0.54	0.81	4	30	80
Pharmatose® 350M NZ**	●●●	●	o	o	o	●●●	●●	●	●	0.54	0.70	5	30	80
Pharmatose® 200M	●●●	●	o	o	●	●●	●●	●	●	0.59	0.86	5	40	120
Pharmatose® 200M NZ**	●●●	●	o	o	●	●●	●●	●	●	0.56	0.72	4	30	110
Pharmatose® 150M	●●●	●	o	o	●●	●	●●	●	●	0.65	0.91	6	60	210
Pharmatose® 130M	●●●	●	o	o	●●	●	●●	●	●	0.70	0.87	9	80	220
Lactochem® super fine powder	●●●	●	o	o	o	●●●	●●●	●	●	0.42	0.68	3	15	40
Lactochem® extra fine powder	●●●	●	o	o	o	●●●	●●	●	●	0.43	0.70	3	20	50
Lactochem® fine powder	●●●	●	o	o	●	●●	●●	●	●	0.54	0.83	4	30	100
Lactochem® regular powder***	●●●	●	o	o	●●	●	●●	●	●	0.67	0.93	9	70	170
Lactochem® powder	●●●	●	o	o	●●	●	●●	●	●	0.57	0.87	6	50	140
Lactochem® coarse powder	●●●	●	o	o	●●	●	●●	●	●	0.70	0.94	15	100	210
Sieved														
Pharmatose® 125M	o	o	●	●●	●●	●	●	●	●	0.72	0.85	30	80	140
Pharmatose® 110M	o	o	●	●●	●●	o	●	●	●	0.75	0.87	45	130	230
Pharmatose® 100M	o	o	●	●●	●●	o	●	●	●	0.76	0.88	50	150	250
Pharmatose® 90M	o	o	●	●●	●●●	o	●	●	●	0.75	0.87	50	170	280
Pharmatose® 80M	o	o	●	●●	●●●	o	●	●	●	0.79	0.90	70	250	390
Pharmatose® 70M	o	o	●	●●	●●●	o	●	●	●	0.83	0.92	75	220	410
Pharmatose® 60M	o	o	●	●●	●●●	o	●	●	●	0.82	0.89	135	280	470
Pharmatose® 50M	o	o	●	●●	●●●	o	●	●	●	0.72	0.80	235	360	490
Lactochem® extra fine crystals	o	o	●	●●	●●	●	●	●	●	0.73	0.90	35	140	230
Lactochem® fine crystals	o	o	●	●●	●●	o	●	●	●	0.77	0.91	40	160	260
Lactochem® crystals	o	o	●	●●	●●●	o	●	●	●	0.75	0.86	50	140	220
Lactochem® coarse crystals	o	o	●	●●	●●●	o	●	●	●	0.73	0.82	60	150	240
Anhydrous														
Lactopress® anhydrous fine powder	●	●●●	●●	●	●	●●	●●●	●	●●●	0.50	0.80	3	20	70
Lactopress® anhydrous powder	●	●●●	●●	●	●	●●	●●●	●	●●●	0.57	0.89	4	50	160
Lactopress® anhydrous	●	●●●	●●●	●●●	●●	●●	●●●	●	●●●	0.60	0.85	15	130	340
SuperTab® 21AN	●	●●●	●●●	●●●	●●	●●	●●●	●	●●●	0.72	0.91	15	170	340
SuperTab® 22AN	●	●●●	●●●	●●●	●●●	●●	●●●	●	●●●	0.68	0.80	70	220	380
SuperTab® 24AN	●	●	●●●	●●●	●●●	●●●	●●●	●●●	●●●	0.49	0.60	40	140	300
Spray Dried														
Lactopress® spray dried	●	●	●●●	●●	●●●	●	●	●●	●●	0.64	0.73	55	130	220
Lactopress® spray dried 250	●	●	●●●	●●	●●●	●	●	●●	●●	0.64	0.74	45	110	200
SuperTab® 11SD	●	●	●●●	●●	●●●	●	●	●●	●●	0.60	0.72	50	120	220
SuperTab® 11SD NZ**	●	●	●●●	●●	●●●	●	●	●●	●●	0.62	0.69	50	140	240
SuperTab® 14SD	●	●	●●●	●●	●●●	●●	●	●●	●●	0.62	0.72	50	130	240
SuperTab® 50ODT	●	●	●●●	●●	●●●	●●	●	●●●	●●	0.64	0.73	55	130	220
Granulated														
Lactopress® Granulated	●	o	●●●	●●●	●●●	●	●●●	●●●	●●	0.56	0.69	55	160	300
SuperTab® 30GR	●	o	●●●	●●●	●●●	●	●●●	●●●	●●	0.55	0.67	40	140	300
SuperTab® 40LL	●	●●	●●●	●●●	●●●	●●●	●●●	●	●●	0.54	0.66	80	180	350
MCC														
Pharmacel® 101	●●●	●●	●●	●	●	●●●	●	●●●	o	0.28	0.44	25	60	120
Pharmacel® 102	●●	●●	●●●	●●	●●	●●●	●	●●●	o	0.33	0.46	40	90	180
Pharmacel® 112	●	●●	●●●	●●	●●	●●●	●	●●●	●●	0.31	0.44	40	90	180
Pharmacel® sMCC90	●●	●●	●●●	●●	●●●	●●●	●●	●●●	o	0.31	0.47	35	110	270
Starches														
Prejel PA5 PH	●●●	●	●	●●	●	o	●	●	o	0.59	0.79	15	55	130
Solani Amylum	●●●	●	●	●●	●	o	●	●●	o	0.71	0.90	20	40	65
Superdisintegrants														
Primellose	●●●	●●●	●●●	●	●●	o	o	●●●	o	0.55	0.74	20	50	110
Primojel	●●●	●●●	●●●	●	●●	o	o	●●●	o	0.79	0.96	20	40	70

* = For sachet filling we recommend (coarse) sieved grades
** = New Zealand
*** = Low endotoxin grade available

Overview of all **DFE Pharma OSD grades**

		Dosage form				Process technics		
	Grade	Tablets	Capsules	Sachets	Spheres	Direct compression	Wet granulation	Dry granulation
Sieved lactose	Pharmatose® 50M		•	•				
	Pharmatose® 60M		•	•				
	Pharmatose® 70M		•	•				
	Pharmatose® 80M		•	•				
	Pharmatose® 90M		•	•				
	Pharmatose® 100M	•	•	•				
	Pharmatose® 110M	•	•	•				
	Pharmatose® 125M	•	•	•				
	Lactochem® Coarse Crystals		•	•				
	Lactochem® Crystals	•	•	•				
	Lactochem® Coarse Powder	•			•		•	
	Lactochem® Regular Powder	•			•		•	
Milled lactose	Pharmatose® 130M	•			•		•	
	Pharmatose® 150M	•			•		•	
	Pharmatose® 200M	•			•		•	
	Pharmatose® 350M	•			•		•	
	Pharmatose® 450M	•			•		•	
	Lactochem® Coarse Powder	•	•	•	•		•	
	Lactochem® Regular Powder	•			•		•	
	Lactochem® Powder	•			•		•	
	Lactochem® Fine Powder	•			•		•	
	Lactochem® Extra Fine Powder	•			•		•	
	Lactochem® Super Fine Powder	•			•		•	
Spray dried lactose	SuperTab® 11SD	•	•	•		•		
	SuperTab® 14SD	•	•	•		•		
	Lactopress® Spray Dried	•	•	•		•		
	Lactopress® Spray Dried 250	•	•	•		•		
Anhydrous lactose	SuperTab® 21AN	•	•			•		•
	SuperTab® 22AN	•	•			•		•
	SuperTab® 24AN*	•	•			•		
	Lactopress® Anhydrous	•	•	•		•		•
	Lactopress® Anhydrous Powder	•	•	•		•		•
	Lactopress® Anhydrous Fine Powder	•	•			•		•
Granulated lactose	SuperTab® 30GR	•	•	•		•		
	SuperTab® 24AN*	•	•			•		
	Lactopress® Granulated	•	•	•		•		

		Dosage form				Process technics		
	Grade	Tablets	Capsules	Sachets	Spheres	Direct compression	Wet granulation	Dry granulation
ODT lactose	SuperTab® 50 ODT	•				•		
Specialty lactose	Lactochem® Regular Powder Low Endotoxin	•			•		•	
MCC	Pharmacel® 101	•	•		•		•	•
	Pharmacel® 102	•	•			•		
	Pharmacel® 112	•	•			•		•
Superdisintegrants	Primojel®	•	•		•	•	•	•
	Primellose®	•	•		•	•	•	•
Starch	Solani Amylum	•	•		•		•	
	Prejel PA5 PH	•	•				•	
Co-processed	SuperTab® 40LL	•				•		